

Shared evidence • The water-path case file

The water-path detectives

Name _____ Date _____

Records C-F are constructed teaching observations. Real dye timing varies; use the evidence supplied today.

A • Rooted-plant model

Soil water enters roots, then travels inside the stem to leaves. Some water leaves as vapour.

C • After 2 hours

Constructed observation: red lines appear inside the dye shoot's adult-cut stem section.

E • Plain-water comparison

Constructed observation: no red dye colour in the control stem or petals.

B • Setup

Two similar cut white-flower shoots: one in red-coloured water, one in plain water. Neither has roots.

D • After 24 hours

Constructed observation: some petals show red colour. Exact timing varies in real plants.

F • Water-level check

A separate open cup without a plant also loses some water. Evaporation is possible.

W1. Put setup, internal colour and petal colour in order.

W2. Which part of the full rooted-plant route was not tested by B?

Route 1 • Build the water route

Word bank: soil / roots / stem / leaves / inside.

S1. Complete the route: soil → ____ → ____ → leaves.

S2. The red paths in C are...

- A. Inside the stem section.
- B. Only drawn on its outside.

S3. What was different about the water in the comparison shoot?

S4. Did the cut shoots have roots?

- A. Yes
- B. No

S5. State one thing the dye evidence supports.

Route 2 • Write a detective explanation

Use the observation and explain what it supports.

M1. Draw or describe the full water route in a rooted plant.

M2. Explain how C and D support transport in the cut shoot.

M3. Explain why E is a useful comparison.

M4. State one thing B did not test and one reason F matters.

Route 3 • Challenge the overclaim

A report says: “All water travelled up the outside. We proved root uptake, and every drop missing from the cup entered the plant.”

T1. Use C to challenge the outside-only claim.

T2. Use B to explain why direct root-uptake proof is missing.

T3. Use F to explain an alternative cause of water loss.

T4. Write a careful two-sentence conclusion that keeps the useful evidence.

Exit and reflection

Point, draw, write or tell an adult.

E1. Which part takes in water from soil?

E2. Does water travel through internal stem pathways?

E3. Did the cut-shoot experiment contain roots?

R1. Which idea did you improve or defend? What evidence helped?

R2. What useful question would you investigate next?
